

6032D Electric Vehicles & Traction

POLY PMNA - Polytechnic Study Hub

Semester 6 | Program Elective | Credits 4 | Periods 60

Downloadable handbook generated from lessons-6032D.html

Core areas: EV powertrain, vehicle motion, electric traction, speed time curve, BLDC, PMSM, batteries, hybrid storage, fuel cells, charging standards and e-mobility.

Course Overview and Roadmap

Objectives: express EV relevance, illustrate electric traction and EV motors, outline energy storage options and identify charging infrastructure requirements.

CO1: Outline basic concepts of EVs and match power train needs. CO2: Explain electric traction and EV motor applications. CO3: Outline importance of hybridization of energy storage. CO4: Summarize public charging infrastructure and e-mobility.

Roadmap: M1 vehicle classification, motion resistance and EV/HEV/PHEV powertrain. M2 electric traction, traction motors, speed-time curve, BLDC and PMSM. M3 batteries, ultra-capacitors, flywheels, fuel cells and hybrid storage. M4 HEV control, chargers, connectors, V2G/G2V/V2B/V2H, public charging and e-mobility challenges.

Module Chapters

Module 1: EVs, HEVs and PHEVs are compared by technical, economical and environmental aspects. Vehicle movement depends on rolling resistance, aerodynamic drag, grade resistance, adhesion and tractive effort. IC powertrain uses clutch, gearbox and differential; EV powertrain uses battery, inverter/controller, motor and drive train.

Module 2: Electric traction includes supply system, locomotive layout, traction motors, speed-time curve and train movement terms. Traction motors include DC series, single phase AC series and three phase induction motors. Special EV motors include BLDC and PMSM with electronic/synchronous control.

Module 3: Battery parameters include specific power, energy efficiency, thermodynamic voltage and selection criteria. Li-ion and Li-P batteries are common EV choices. Ultra capacitors, flywheels and hydrogen fuel cells support high power or range needs.

Module 4: HEV control coordinates engine, motor, battery, inverter and charger. EV charging covers SAE J1772 modes, CCS, Bharat EV AC/DC, safety, connectors, public charging requirements and e-mobility challenges.

Formula Bank, Practice and Answers

Formula bank: total tractive effort equals rolling, aerodynamic, grade and acceleration resistances. Average speed equals distance divided by actual running time. Scheduled speed equals distance divided by total time including stops. Battery energy equals $V \times Ah$.

Solved example: 320 V, 100 Ah battery has 32000 Wh = 32 kWh. With 6 kW charger, ideal charging time is $32 / 6 = 5.33$ hours.

Expected questions: compare EV/HEV/PHEV, explain vehicle resistance, speed-time curve, BLDC, PMSM, storage hybridization, fuel cells, charging modes, connectors and public charging infrastructure.

Model answer guide: include diagrams, formulas, units, torque-speed keywords, safety, communication, connector rating and grid interaction. Do not omit BMS and controller.